Chunlin Song

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State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resources and Hydropower, Sichuan University, No.24 South Section 1, Yihuan Road, Chengdu, China

Research Interest

Riverine carbon cycle; Lake carbon cycle; aquatic greenhouse gases; permafrost hydrology; permafrost carbon cycle; watershed ecohydrology

Professional Experience

State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resources and Hydropower, Sichuan University *Sept. 2023 - Present*
Associate Professor

State Key Laboratory of Hydraulics and Mountain River Engineering, College of Water Resources and Hydropower, Sichuan University *Jun. 2020 - Aug. 2023*
Tenure-track Associate Professor

Institute of Mountain Hazards and Environment, Chinese Academy of Sciences *Jul. 2019 - Jun. 2020*
Assistant Researcher

Education

Institute of Mountain Hazards and Environment, Chinese Academy of Sciences & University of Chinese Academy of Sciences *Sept. 2013 - Jun. 2019*
Combined Master's and PhD program in Physical Geography, advisor: Professor Genxu Wang
Dissertation: Riverine Carbon Export Processes of the Yangtze River Source Region

Yale School of Forestry and Environmental Studies *Nov. 2017 - Jan. 2019*
Visiting PhD student, river biogeochemistry, advisor: Professor Peter A. Raymond

College of Water Resources and Hydropower, Sichuan University *Sept. 2009 - Jun. 2013*
Bachelor of Engineering in Hydrology and Water Resources Engineering

Publications

Papers not yet published

- **Chunlin Song**, Genxu Wang, Cristian Gudasz, R Iestyn Woolway, Hongyang Chen, Yang Li, Jan Karlsson. Lake size shapes the temperature dependence of methane emissions in cryosphere lakes. (In Revision).
- **Chunlin Song***, Yuhao Li, Genxu Wang, Martin R. Kurek, Travis W. Drake, Cuicui Mu, Chao Liu, Xiangyang Sun, Qiongwei Gu, Shan Lin, Jiapei Ma, Jinlong Li, Robert G. M. Spencer. Seasonal aliphatic enrichment drives high DOC loss through direct photo-biological degradation over photo-priming in permafrost streams. (In Review).
- Hongqiao Chen, **Chunlin Song***, Genxu Wang*, Penghui Shi, Hanrui Zhao, Ying Wang, Hengyu Pan, Jan Karlsson. Inland waters weaken restoration-driven carbon sink gains in the Upper Yangtze River Basin. (In Review).

Updated August 2026. More info: **Google Scholar; ResearchGate; ORCiD; Website; X; CV in Chinese**

Peer-reviewed papers as a first author (# I led the work) **or corresponding author** (* I conceived and designed these studies and provided intellectual input and funding and resource support; my students conducted the research and wrote the paper under my supervision)

26. Hongqiao Chen, **Chunlin Song***, Genxu Wang, Hengyu Pan, Liwei Zhang, Yuanzhi Yao, Xincong Liu, Xiangyu Zheng, Lang Huang, Huanyu Wang, Xuqin Wu, Yixuan Gao, Jan Karlsson (2026). Recent increase of greenhouse gas emissions from cryosphere-influenced reservoirs under climate change. *Environmental Science & Technology*, 60 (34): 24101-24112. https://doi.org/10.1021/acs.est.6c08337

25. **Chunlin Song***, Genxu Wang, Jiapei Ma (2026). Broadening the discourse on mitigating carbon emissions from thawing permafrost. *Science Bulletin*, 71(17):4338-4340. https://doi.org/10.1016/j.scib.2026.04.054.

24. Jiali Ding, Genxu Wang, Yuhao Li, Qiongwei Gu, Jinlong Li, **Chunlin Song*** (2026). Negligible greenhouse gas emissions from the exposed riverbeds in permafrost region revealed by year-round observations. *Inland Waters*, https://doi.org/10.1080/20442041.2026.2679574.

23. Jiapei Ma, Genxu Wang, **Chunlin Song***, Shouqin Sun, Jinlong Li, Zhao Deng, Wendong Kang, Zhuoxuan Xia, Lin Liu (2026). Integrating time-series analysis and deep learning methods to reconstruct the long-term retrogressive thaw slumps dynamics in the Qinghai-Tibet Plateau. *GIScience Remote Sensing*, 63(1), 2617753. https://doi.org/10.1080/15481603.2026.2617753

22. Yang Li, Genxu Wang, Shouqin Sun, Ruiying Chang, Tao Wang, Zhaoyong Hu, Fenglin Xu, Yi Yang, **Chunlin Song*** (2025). Net carbon dioxide sequestration by large alkaline lakes dominates the carbon exchange of Qinghai-Tibet Plateau lakes. *Communications Earth Environment*, 6, 952. https://doi.org/10.1038/s43247-025-02884-z.

21. Ying Wang, Yuhao Li, Genxu Wang, Yang Li, Xiangyang Sun, Wenzhi Wang, **Chunlin Song*** (2025). Diverse altitudinal patterns and drivers of greenhouse gas dynamics in Southwest China alpine streams and rivers. *Journal of Geophysical Research: Biogeosciences*, 130(2), e2024JG008458. https://doi.org/10.1029/2024JG008458

20. Yuhao Li, Genxu Wang, Wenzhi Wang, Xiangyang Sun, Yang Li, Jinwang Xiao, Wentian Xie, Jiali Ding, **Chunlin Song*** (2025). Ultraviolet radiation stimulates the degradability of groundwater–fed DOC during the baseflow period of streams on the Qinghai-Tibet Plateau permafrost region. *Journal of Hydrology*, 653, 132751. https://doi.org/10.1016/j.jhydrol.2025.132751

19. **Chunlin Song**[#*], Shaoda Liu[#], Genxu Wang[#*], Liwei Zhang, Judith A. Rosentreter, Gang Zhao, Xiangyang Sun, Yuanzhi Yao, Cuicui Mu, Shouqin Sun, Zhaoyong Hu, Shan Lin, Juying Sun, Yang Li, Ying Wang, Yuhao Li, Peter A. Raymond, Jan Karlsson (2024). Inland water greenhouse gas emissions offset the terrestrial carbon sink in the northern cryosphere. *Science Advances*, 10(39), eadp0024. https://doi.org/10.1126/sciadv.adp0024

18. Yang Li, Genxu Wang*, Shouqin Sun, Shan Lin, Peng Huang, Jinwang Xiao, Linmao Guo, Jinlong Li, **Chunlin Song*** (2024). Methane emissions from the Qinghai-Tibet Plateau ponds and lakes: Roles of ice thaw and vegetation zone. *Global Biogeochemical Cycles*, 38(4), e2024GB008106. https://doi.org/10.1029/2024GB008106

17. Jinlong Li, Genxu Wang*, **Chunlin Song***, Shouqin Sun, Jiapei Ma, Ying Wang, Linmao Guo, Dongfeng Li (2024). Recent Intensified Erosion and Massive Sediment Deposition in the Tibetan Plateau Rivers. *Nature Communications* 15, 722. https://doi.org/10.1038/s41467-024-44982-0

16. Ying Wang, Genxu Wang, Xiangyang Sun, Jinlong Li, **Chunlin Song*** (2024). Spatiotemporal variability of organic carbon in streams and rivers of the Northern Hemisphere cryosphere. *Science of The Total Environment*, 167370. https://doi.org/10.1016/j.scitotenv.2023.167370

15. Jinlong Li, Genxu Wang*, Kai Li, Yang Li, Linmao Guo, **Chunlin Song*** (2023). Impacts of climate change

and freeze-thaw cycles on water and sediment fluxes in the headwater region of the Yangtze River, Qinghai-Tibet Plateau. *Catena*, 227C, 107112. https://doi.org/10.1016/j.catena.2023.107112

14. **Chunlin Song**, Genxu Wang*, Xiangyang Sun, Yang Li, Silu Ye, Zhaoyong Hu, Juying Sun, Shan Lin (2023). Riverine CO_2 variations in permafrost catchments of the Yangtze River source region: Hot spots and hot moments. *Science of The Total Environment*, 863, 160948, https://doi.org/10.1016/j.scitotenv.2022.160948

13. Zhiwei Wang, Shouqin Sun, Genxu Wang*, **Chunlin Song*** (2023). Determination of low-flow components in alpine permafrost rivers. *Journal of Hydrology*, 617, 128886, https://doi.org/10.1016/j.jhydrol.2022.128886

12. Zhiwei Wang, Shouqin Sun, **Chunlin Song***, Genxu Wang*, Shan Lin, Silu Ye (2022). Variation characteristics of high flows and their responses to climate change in permafrost regions on the Qinghai-Tibet Plateau, China. *Journal of Cleaner Production*, 376, 134369, https://doi.org/10.1016/j.jclepro.2022.134369

11. **Chunlin Song** and Genxu Wang* (2021). Land carbon sink of the Tibetan Plateau may be overestimated without accounting the aquatic carbon export. *Proceedings of the National Academy of Sciences of the United States of America*, 118(46), e2114694118, https://doi.org/10.1073/pnas.2114694118 (Letter)

10. **Chunlin Song**, Genxu Wang*, Xiangyang Sun, Zhaoyong Hu (2021). River runoff components change variably and respond differently to climate change in the Eurasian Arctic and Qinghai-Tibet Plateau permafrost regions. *Journal of Hydrology*, 601, 126653, https://doi.org/10.1016/j.jhydrol.2021.126653

9. **Chunlin Song**, Genxu Wang*, Zhaoyong Hu, Kewei Huang, Tao Zhang, Xiaopeng Chen, Yang Li (2020). Net ecosystem carbon budget of a grassland ecosystem in central Qinghai-Tibet Plateau: integrating terrestrial and aquatic carbon fluxes at catchment scale. *Agricultural and Forest Meteorology*, 290, 108021, https://doi.org/10.1016/j.agrformet.2020.108021

8. **Chunlin Song**, Genxu Wang*, Negar Haghipour, Peter A. Raymond (2020). Warming and monsoonal climate lead to a large export of millennial-aged carbon from permafrost catchments of the Qinghai-Tibet Plateau. *Environmental Research Letters*, 15(7): 074012. https://doi.org/10.1088/1748-9326/ab83ac

7. **Chunlin Song**, Genxu Wang*, Tianxu Mao, Kewei Huang, Xiangyang Sun, Zhaoyong Hu, Ruiying Chang, Xiaopeng Chen, Peter A. Raymond (2020). Spatiotemporal variability and sources of DIC in permafrost catchments of the Yangtze River source region: insights from stable carbon isotope and water chemistry. *Water Resources Research*, 56(1): e2019WR025343. https://doi.org/10.1029/2019WR025343

6. **Chunlin Song**, Genxu Wang*, Tianxu Mao, Junchen Dai, Daqing Yang (2020). Linkage between permafrost distribution and river runoff changes across the Arctic and the Tibetan Plateau. *Science China Earth Sciences*, 63(2): 292-302. https://doi.org/10.1007/s11430-018-9383-6

5. **Chunlin Song**, Genxu Wang*, Tianxu Mao, Xiaopeng Chen, Kewei Huang, Xiangyang Sun, Zhaoyong Hu (2019). Importance of active layer freeze-thaw cycles on the riverine dissolved carbon export on the Qinghai-Tibet Plateau permafrost region. *PeerJ*, 7:e7146. https://doi.org/10.7717/peerj.7146

4. **Chunlin Song**, Genxu Wang*, Guangsheng Liu, Tianxu Mao, Xiangyang Sun, Xiaopeng Chen (2017). Stable isotope variations of precipitation and streamflow reveal the young water fraction of a permafrost watershed. *Hydrological Processes*, 31(4), 935-947. https://doi.org/10.1002/hyp.11077

3. **Chunlin Song**, Genxu Wang*, Xiangyang Sun, Ruiying Chang, Tianxu Mao (2016). Control factors and scale analysis of annual river water, sediments and carbon transport in China. *Scientific Reports*, 6:25963. https://doi.org/10.1038/srep25963

2. **Chunlin Song**, Xiangyang Sun, Genxu Wang* (2015). A study on precipitation stable isotopes characteristics and vapor sources of the subalpine Gongga Mountain, China. *Resources and Environment in the Yangtze Basin*, 24(11), 1860-1869. (in Chinese with English Abstract)

1. **Chunlin Song**, Xiangyang Sun, Genxu Wang* (2015). A review on carbon and water interactions of forest ecosystem and its impact factors. *Chinese Journal of Applied Ecology*, 26(9), 2891-2902. (in Chinese with English Abstract)

Peer-reviewed Papers as a co-author

30. Judith A. Rosentreter, Alberto V. Borges, Bridget R. Deemer, Meredith A. Holgerson, Shaoda Liu, **Chunlin Song**, John Melack, Peter A. Raymond, Carlos M. Duarte, George H. Allen, David Olefeldt, Benjamin Poulter, Tom I. Battin Bradley D. Eyre (2021). Half of global methane emissions come from highly variable aquatic ecosystem sources. *Nature Geoscience*, 14, 225-230. https://doi.org/10.1038/s41561-021-00715-2

29. Cuicui Mu, Kun Li, Shaoda Liu, Yuguo Wei, Mei Mu, Xuexue Shang, Fumei Liu, Chunling Zhang, Hebin Liu, Tanguang Gao (2025). Recent intensified riverine CO_2 emission across the Northern Hemisphere permafrost region. *Nature Communications*, 16(1), 3616.

28. Cuicui Mu, Pengsi Lei, Mei Mu, Chunling Zhang, Zhensong Zhou, Jinyue Song, Yunjie Jia, Chenyan Fan, Xiaoqing Peng, Guofei Zhang (2025). Methane emissions from thermokarst lakes must emphasize the ice-melting impact on the Tibetan Plateau. *Nature Communications*, 16(1), 2404.

27. Hanrui Zhao, Shouqin Sun, **Chunlin Song**, Shan Lin, Zhaoyong Hu, Xianhong Meng, Jinlong Li, Linmao Guo, Genxu Wang (2026). Freeze-thaw-driven soil moisture return significantly contributes to spring phenology on the warming Qinghai-Tibet Plateau. *Nature Communications*, 17, 3891.

26. Jinlong Li, Genxu Wang, Shouqin Sun, Jiapei Ma, Linmao Guo, **Chunlin Song**, Shan Lin (2025). Mapping and reconstruct suspended sediment dynamics (1986-2021) in the source region of the Yangtze River, Qinghai-Tibet Plateau using Google Earth Engine. *Remote Sensing of Environment*, 317, 114533.

25. Nan Jiang, Guoping Tang, Guotao Cui, Haicheng Zhang, Linwei Zeng, Xiaobin Li, **Chunlin Song**, Yuheng Yang, Yiguang Zou, Xixi Lu (2026). Examining the seasonal-transition effect on dissolved carbon sources and exports in a subtropical headwater catchment. *Water Resources Research*, 62(7), e2025WR042970.

24. Václava Hazuková, Fredrik Alriksson, Cristian Gudasz, Jan Karlsson, **Chunlin Song**, Matthew J Bogard, David E Butman, Joshua F Dean, Anders Jonsson, Erik Lundin, Suzanne E Tank, Jasmine E Saros (2026). Higher, but more variable, annual CO2 emissions from lakes in drier Arctic landscapes. *Communications Earth & Environment*, 7(1), 238.

23. Jiapei Ma, Genxu Wang, Shouqin Sun, **Chunlin Song**, Jinlong Li, Linmao Guo, Kai Li, Peng Huang, Shan Lin (2025). An enhanced image stacks method for mapping long-term retrogressive thaw slumps in the Tibetan Plateau. *International Journal of Applied Earth Observation and Geoinformation*, 137, 104414.

22. Linmao Guo, Genxu Wang, **Chunlin Song**, Shouqin Sun, Jinlong Li, Kai Li, Peng Huang, Jiapei Ma (2025). Hydrological changes caused by integrated warming, wetting, and greening in permafrost regions of the Qinghai-Tibetan Plateau. *Water Resources Research*, 61(4), e2024WR038465.

21. Jinlong Li, Genxu Wang, Jiapei Ma, Peng Huang, **Chunlin Song**, Shouqin Sun, Linmao Guo, Kai Li, Haoyu Wang, Dongfeng Li (2025). Structural connectivity shaped by cryospheric degradation modulates sediment transport dynamics in High Mountain Asia. *Water Resources Research*, 61(6), e2024WR038654.

20. Jiapei Ma, Genxu Wang, **Chunlin Song**, Dexin Gao, Jinlong Li, Peng Huang, Linmao Guo, Kai Li, Shan Lin, Shouqin Sun (2025). Establishing a robust area-to-volume scaling for Qinghai-Tibetan Plateau Retrogressive Thaw Slumps: A key tool for quantifying mass wasting and carbon release induced by permafrost degradation. *Global and Planetary Change*, 105012.

19. Meizhuang Zhu, Xingxing Kuang, Yuqing Feng, Enze Ma, **Chunlin Song**, Zhaoyong Hu, Xiaoyan Shi, Linfeng Fan, Fang Ji, Tian Zhao, Nima Ci Ren, Chunmiao Zheng (2026). Variations of stable carbon isotope imply sources of DIC in an alpine catchment. *Journal of Hydrology*, 666, 134802.

18. Meizhuang Zhu, Xingxing Kuang, **Chunlin Song**, Yuqing Feng, Qiule He, Yiguang Zou, Hui Zhou, Chunmiao Zheng (2024). Glacier-fed lakes are significant sinks of carbon dioxide in the Southeastern Tibetan Plateau. *Journal of Geophysical Research: Biogeosciences*, 129(4), e2023JG007774.

17. Xiaopeng Chen, Genxu Wang, Na Li, Ruiying Chang, Tao Zhang, Tianxu Mao, **Chunlin Song**, Kewei Huang (2024). Nitrogen dynamics of alpine swamp meadows are less responsive to climate warming than that of alpine meadows. *Science of The Total Environment*, 928, 172446.

16. Shan Lin, Xiangyang Sun, Kewei Huang, **Chunlin Song**, Juying Sun, Shouqin Sun, Genxu Wang, Zhaoyong Hu (2024). The seasonal variability of future evapotranspiration over China during the 21st century. *Science of The Total Environment*, 926, 171816.

15. Zhiwei Wang, Shouqin Sun, Genxu Wang, **Chunlin Song** (2024). Spatial-Temporal Differentiation of Supra- and Sub-Permafrost Groundwater Contributions to River Runoff in the Eurasian Arctic and Qinghai-Tibet Plateau Permafrost Regions. *Water Resources Research*, 60(3), e2023WR035913.

14. Juying Sun, Xiangyang Sun, Genxu Wang, Wenchang Dong, Zhaoyong Hu, Shouqin Sun, Fei Wang, **Chunlin Song**, Shan Lin (2024). Soil water components control plant water uptake along a subalpine elevation gradient on the Eastern Qinghai-Tibet Plateau. *Agricultural and Forest Meteorology*, 345, 109827.

13. Xiangyang Sun, Xinyu Zhang, Genxu Wang, Zhaoyong Hu, **Chunlin Song**, Shan Lin, Juying Sun, Shouqin Sun (2023). An Increasing Effect of Soil Moisture on Semiempirical WaterâĂŘUse Efficiency Models From Wet to Dry Climate Regions. *Journal of Geophysical Research: Biogeosciences*, 128(6), e2022JG007347.

12. Zhaoyong Hu, Shouqin Sun, Xiangyang Sun, Shan Lin, **Chunlin Song**, Genxu Wang (2023). Controlling Factors of the SpatialâĂŘTemporal Fluctuations in Evapotranspiration Along an Elevation Gradient Across Humid Montane Ecosystems. *Water Resources Research*, 59(1), e2022WR033228.

11. Yang Li, Genxu Wang, Haijian Bing, Tao Wang, Kewei Huang, **Chunlin Song**, Xiaopeng Chen, Zhaoyong Hu, Pengfei Rui, Xiaoyan Song, Ruiying Chang (2021). Watershed scale patterns and controlling factors of ecosystem respiration and methane fluxes in a Tibetan alpine grassland. *Agricultural and Forest Meteorology*, 306, 108451. https://doi.org/10.1016/j.agrformet.2021.108451

10. Kewei Huang, Junchen Dai, Genxu Wang, Juan Chang, Yaqiong Lu, **Chunlin Song**, Zhaoyong Hu, Naveed Ahmed, Renzheng Ye (2020). The Impact of Land Surface Temperatures on Suprapermafrost Groundwater on the Central Qinghai-Tibet Plateau. *Hydrological Processes*. https://doi.org/10.1002/hyp.13677

9. Xiangyang Sun, Genxu Wang, Mei Huang, Ruiying Chang, Zhaoyong Hu, **Chunlin Song**, Juying Sun (2020). The asynchronous response of carbon gain and water loss generate spatio-temporal pattern of WUE along elevation gradient in southwest China. *Journal of Hydrology*, 124389. https://doi.org/10.1016/j.jhydrol.2019.124389

8. Xiaoyan Song, Genxu Wang, Fei Ran, Kewei Huang, Juying Sun, **Chunlin Song**. (2020). Soil moisture as a key factor in carbon release from thawing permafrost in a boreal forest. *Geoderma*, 357, 113975. https://doi.org/10.1016/j.geoderma.2019.113975

7. Zhaoyong Hu, Genxu Wang, Xiangyang Sun, Jian Wang, Xiaopeng Chen, **Chunlin Song**, Xiaoyan Song, Shan Lin (2019). Variations in belowground carbon use strategies under different climatic conditions. *Agricultural and Forest Meteorology*, 268, 32-39. https://doi.org/10.1016/j.agrformet.2019.01.005

6. Zhaoyong Hu, Genxu Wang, Xiangyang Sun, Meizhuang Zhu, **Chunlin Song**, Kewei Huang and Xiaopeng Chen (2018). Spatial-Temporal Patterns of Evapotranspiration Along an Elevation Gradient on Mount Gongga, Southwest China. *Water Resources Research*, 54(6), 4180-4192. https://doi.org/10.1029/2018WR022645

5. Xiaopeng Chen, Genxu Wang, Tao Zhang, Tianxu Mao, Da Wei, **Chunlin Song**, Zhaoyong Hu, Kewei Huang (2017). Effects of warming and nitrogen fertilization on GHG flux in an alpine swamp meadow of

a permafrost region. *Science of the Total Environment*, 601, 1389-1399. https://doi.org/10.1016/j.scitotenv.2017.06.028

4. Xiaopeng Chen, Genxu Wang, Kewei Huang, Zhaoyong Hu, **Chunlin Song**, Yiming Liang, Jian Wang, Xiaoyan Song, Shan Lin (2017). The effect of nitrogen deposition rather than warming on carbon flux in alpine meadows depends on precipitation variations. ***Ecological Engineering***, 107, 183-191. https://doi.org/10.1016/j.ecoleng.2017.07.018

3. Xiaoyan Song, Genxu Wang, Fei Ran, Ruiying Chang, **Chunlin Song**, Yao Xiao (2017). Effects of topography and fire on soil CO_2 and CH_4 flux in boreal forest underlain by permafrost in northeast China. *Ecological Engineering*, 106, 35-43. https://doi.org/10.1016/j.ecoleng.2017.05.033

2. Genxu Wang, Tianxu Mao, Juan Chang, **Chunlin Song**, Kewei Huang (2017). Processes of runoff generation operating during the spring and autumn seasons in a permafrost catchment on semi-arid plateaus. *Journal of Hydrology*, 550:307-317. https://doi.org/10.1016/j.jhydrol.2017.05.020

1. Xiaopeng Chen, Genxu Wang, Tao Zhang, Tianxu Mao, Da Wei, Zhaoyong Hu, **Chunlin Song** (2017). Effects of warming and nitrogen fertilization on GHG flux in the permafrost region of an alpine meadow. *Atmospheric environment*, 157, 111-124. https://doi.org/10.1016/j.atmosenv.2017.03.024

Patents

- Chunlin Song, Genxu Wang, Xiangyang Sun. Automatic static chamber for greenhouse gas sampling over water surfaces. Chinese invention patent ZL 201610102693.4. Granted 21/06/2019.
- Yuhao Li, Chunlin Song, Genxu Wang, Jiali Ding, Ying Wang, Yang Li, Jinwang Xiao. Long-term river water quality monitoring device. Chinese invention patent application No. 202311787869.0, 22/12/2023.
- Chunlin Song, Genxu Wang, Xiangyang Sun, Shan Lin, Jiapei Ma. Experimental apparatus for permafrost preservation based on renewable energy and a heat pump. Chinese invention patent application No. 202511332577.7 (18/09/2025).
- Ran Wang, Chunlin Song, Ying Wang, Chen Meng, Yuxin Zhou, Mingrui Yu. NB-IoT-based automatic monitoring device for greenhouse gases over water surfaces. Chinese utility patent ZL 202320954156.8. Granted 10/11/2023.
- Jinlong Li, Genxu Wang, Kai Li, Chunlin Song, Shouqin Sun. Google Earth Engine (GEE)-based method for retrieving suspended sediment in rivers on the Qinghai-Tibet Plateau. Chinese invention patent ZL 202211562758.5. Granted 29/08/2023.

Open Access Datasets and Software

- Chunlin Song, Yang Li, Ying Wang, Yuhao Li (2024). Cryosphere Inland Water Greenhouse Gases Database (CIWD-GHG) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8265740
- Jiali Ding, Chunlin Song (2026). Greenhouse gas emissions from permafrost streams and dry riverbeds of the Fenghuoshan watershed on Qinghai-Tibet Plateau [Data set]. Zenodo. https://doi.org/10.5281/zenodo.20119926
- Ying Wang, Chunlin Song, Genxu Wang (2023). Organic carbon data from the northern hemisphere cryosphere rivers and streams [Data set]. Zenodo. https://doi.org/10.5281/zenodo.19481495
- Chunlin Song (2019). Water chemistry and DIC isotope dataset of the Yangtze River source region [Data set]. Figshare. https://doi.org/10.6084/m9.figshare.9891515
- Chunlin Song (2026). Inland water GHG calculations [Computer software]. GitHub. https://github.com/songchunlin/Inla water-GHG-calculations

Monograph

- WANG Genxu, YI Shuhua. *Ecological process of cryosphere changes and carbon cycle impact* (in Chinese). Beijing: Science Press. 2019. Contributed a chapter.
- WANG Genxu, ZHANG Zhiqiang, LI Xiaoyan. *Introduction to Ecohydrology* (in Chinese). Beijing: Science Press. 2019. Contributed a chapter.

Presentations

Oral Presentations

- **Chunlin Song**, Genxu Wang, Cristian Gudasz, R Iestyn Woolway, Hongyang Chen, Yang Li, Jan Karlsson. Lake size shapes the temperature dependence of methane emissions in cryosphere lakes, EGU General Assembly 2026, Vienna, Austria. 05/08/2026.
- **Chunlin Song**. Carbon transport and transformation in cold region water bodies and greenhouse gas emissions, Lanzhou University, Lanzhou, China. 04/06/2026. (Invited)
- **Chunlin Song**. Greenhouse gas emissions from inland waters across the Third Pole and the pan-cryosphere, Carbon Neutrality conference hosted by the Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China. 03/28/2026. (Invited)
- **Chunlin Song**. Aquatic Carbon Cycling and Greenhouse Gas Emissions: From the Third Pole to the pan-Cryosphere. CAS and UNESCO 2025 International Workshop on the Frontiers in Ecohydrology, Chongqing, China. 10/20/2025. (Invited)
- **Chunlin Song**. Greenhouse gas emissions from northern cryosphere inland waters. The IAMAS-IACS-IAPSO Joint Assembly 2025 (BACO-25), Busan, South Korea. 07/21/2025.
- **Chunlin Song**. Carbon Cycling and Greenhouse Gas Emissions in Cold Region Water Bodies. The 3rd Chinese Conference of Cryospheric Science. Ling Shui, Hainan, 03/20/2025 (Invited)
- **Chunlin Song**. Research on greenhouse gas emissions from cryosphere rivers and lakes in the Northern Hemisphere. The 3rd Cryosphere Science Youth Forum, Chengdu, China. 04/13/2024.
- **Chunlin Song**, Genxu Wang. Riverine carbon content, fluxes, sources, and impacts in the Qinghai-Tibet Plateau permafrost watersheds. 28th IUGG General Assembly, Berlin, Gemany. 07/13/2023.
- **Chunlin Song**. Carbon export from permafrost catchments of the Qinghai-Tibet Plateau. 2022 International Symposium on Ecohydraulics. Online Meeting. 10/11/2022.
- **Chunlin Song**, Genxu Wang, Negar Haghipour, Peter A. Raymond. Millennial-aged carbon export from permafrost catchments of the Qinghai-Tibet Plateau. Cryosphere Forum 2021: Status of research on changing permafrost and associated impacts in the Hindu Kush Himalaya. Online Meeting. 09/23/2021.
- **Chunlin Song**. The dynamics, sources and effects of carbon export from rivers in permafrost regions of the Qinghai-Tibet Plateau. Young Scientists Forum of the Chinese Academy of Sciences. Chengdu, China, 09/08/2021.
- **Chunlin Song**. The influence of permafrost changes in the Qinghai-Tibet Plateau on the riverine export of dissolved carbon. The 7th Young Scientist Forum of Earth Science. Guiyang, China, 07/11/2021.
- **Chunlin Song**. Basin-scale Net Ecosystem Carbon Budget with the Consideration of Aquatic Carbon Flux: A Case Study of Fenghuoshan Watershed in Qinghai-Tibet Plateau. The 2nd China Ecohydrology Forum. Changsha, China, 07/03/2021.
- **Chunlin Song**. The patterns and mechanisms of carbon export from permafrost rivers on the Qinghai-Tibet Plateau. The 19th China Ecology Conference. Online, 11/21/2020.

- **Chunlin Song**. Carbon export patterns and mechanisms in typical permafrost draining rivers in of the Qinghai-Tibet Plateau permafrost region. Mountain Science Young Scholars Development Forum, Chengdu, China. 09/20/2019.
- Peter A. Raymond, **Chunlin Song**, Shaoda Liu, George H. Allen. Stream and River Methane Emissions. AGU Fall Meeting 2018, Washington, D.C., USA. 12/14/2018.
- **Chunlin Song**, Genxu Wang, Tianxu Mao. Seasonal riverine export of dissolved carbon affected by active layer freeze-thaw cycles in headwater streams of the Qinghai-Tibet Plateau permafrost region. AGU Fall Meeting 2017, New Orleans, USA. 12/13/2017.
- Genxu Wang, **Chunlin Song** (speaker). Mechanism of the surface runoff processes of a permafrost watershed in the Qinghai-Tibet plateau. The 2nd Asian Conference on Permafrost, Sapporo, Japan. 07/03/2017.
- **Chunlin Song**, Tianxu Mao. Carbon export and its control factors of Yangtze River source region. Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu, China. 11/13/2016.
- **Chunlin Song**. Invited to give a presentation about doctoral life to undergraduates at the mountain science summer school. Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu, China. 07/21/2016.
- **Chunlin Song**. Theory and application of Meta-analysis. Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, Chengdu, China. 05/03/2016.

Funded Grants

PI: Sichuan Science and Technology Program Key Project, 200,000 CNY 2026.01 - 2028.12
Study on net ecosystem carbon balance at the the Upper Yangtze River basin by integrating terrestrial and aquatic carbon fluxes

PI: National Natural Science Foundation of China General Program, 450,000 CNY 2025.01 - 2028.12
The diel rhythm and driving mechanisms of riverine carbon cycle in the Qinghai-Tibet Plateau permafrost region

PI: Sichuan University "From 0 to 1" Innovation Research Project, 140,000 CNY 2024.01 - 2025.12
Fine characterization and evolution mechanism on water and sediment dynamic processes in the cold region watersheds of the Qinghai-Tibet Plateau

PI: Sub-Project of National Key Research and Development Program of China, 580,000 CNY 2022.12 - 2025.11
The influence and simulation of energy and water phase changes on the processes of flow generation and yield in the source area of the Yangtze River and the Yellow River

PI: National Natural Science Foundation of China Young Scientists Fund, 300,000 CNY 2022.01 - 2024.12
Carbon dioxide emissions from water-air interface of permafrost rivers and the effects of freeze-thaw processes in the Qinghai-Tibet Plateau

PI: Sichuan Science and Technology Program Young Scientists Fund, 100,000 CNY 2022.01 - 2023.12
Characteristics and driving factors of methane emission from alpine rivers and streams along altitude gradients in Hailuogou, Gongga Mountain

PI: Sichuan University Startup Funding for Research, 300,000 CNY 2020 - 2023
Key Processes of River Carbon Cycle in Permafrost Regions of the Qinghai-Tibet Plateau and Their Effects on Watershed Carbon Balance

PI: Sub-Project of Sanjiangyuan National Park Joint Research Program of the Chinese Academy of Sciences and The People's Government of Qinghai Province, 400,000 CNY 2020 - 2022

1.4 Water and heat transport and coupling processes of the SPAC system of typical underlying surface in the three-rivers headwater region & 4.2 Safety Protection Strategy of the water system of three-rivers headwater region

Co-PI: Sub-Project of the Second Tibetan Plateau Scientific Expedition and Research Program (STEP), 800,000 CNY 2019 - 2022
The Eco-hydrological Process and Water Conservation of Typical Wetland Ecosystem in the Qinghai-Tibet Plateau

Teaching & Advising Experience

Teaching

Joint-teaching graduate course: *Ecohydrology* Fall 2021 - present

Advising Experience

Supervised 4 Master's students: Silu Ye (2020 – 2023), Ying Wang (2021 – 2024), Yuhao Li (2022 – 2025), Jiali Ding (2023 – 2026)

Co-supervised 4 PhD students: Yang Li (2020 – 2025), Jinlong Li (2021 – 2025), Zhiwei Wang (2020 – 2024), Jiapei Ma (2022 – 2026)

Supervised 3 undergraduate research groups led by Ran Wang, Liangling Jiang, and Dengbang Zhou

Public Outreach

- Invited Podcast interview in Hydro90 EP25: A Guide to Unlocking the Transition from Student to Junior Faculty (in Chinese). 09/16/2025.
- Invited talk: The secret breathing of the cryosphere inland waters, undergraduate seminar, Sichuan University, China. 10/26/2025.
- Invited talk: Studying abroad, undergraduate seminar, Sichuan University, China. 12/18/2020.
- Invited talk: How I survived my PhD, postgraduate seminar, Institute of Mountain Hazards and Environment, Chinese Academy of Sciences, China. 15/10/2020.

Awards & Honors

Top 5% of Highly Cited Scholars in China in 2025

Wiley China Excellent Author Program, 2025

Shi Yafeng Prize for Young Scientists in Cryosphere and Environment, 2025

Outstanding Instructor Award in the National College Student Water Conservancy Innovation Design Competition in 2023

Outstanding Doctoral Dissertation of the Chinese Academy of Sciences, 2020

China Scholarship Council Fellowship for Joint Ph.D. Training Program, 2017

Bronze Prize of ACOP 2017 Photo Contest

Pacemaker of Merit Student Award, 2017 (1%)

National Scholarship for Doctoral Students, 2016 (6%)

First Class Prize of the IMHE Academic Award, 2016 (10%)

Second Class Prize of the Director Scholarship of the Chengdu Branch of the Chinese Academy of Sciences, 2015 (1.7%)

National Scholarship for Master Students, 2014 (3%)

First Class Prize of the IMHE Academic Award, 2014 (6%)

Merit Student Award of the University of Chinese Academy of Sciences, 2014, 2015, & 2016 (15%)

Sichuan University Excellent Undergraduate Thesis Award, 2013 (5%)

Professional Engagement Activities

Field Trips

Qinghai-Tibet Plateau major rivers (Yangtze, Yellow, Lancang, etc.), Gongga Mountain, Southwest China rivers (Jinsha, Min, Dadu, Yalong, Jialing, etc.), Connecticut River (New England region, USA)

Academic Services

Journal Peer Reviews

Advances in Climate Change Research
Advances in Water Resources
Catena
Communications Earth & Environment
Earth's Future
Earth and Space Science
Earth System Science Data
Environmental Research Letters
Environmental Reviews
Environmental Science & Technology
Frontiers in Earth Science
Frontiers in Environmental Science
Geoderma
Global Biogeochemical Cycles
Global Change Biology
Hydrology and Earth System Sciences
Hydrological Processes
Inland Waters
Journal of Geophysical Research – Atmosphere
Journal of Geophysical Research – Biogeosciences
Journal of Hydrology
Journal of Hydrology: Regional Studies
Journal of Mountain Science
Journal of Soils and Sediments
Limnology and Oceanography
Limnology and Oceanography Letters
Limnology and Oceanography: Methods
Quaternary International
Remote Sensing
Science Advances
Science Bulletin
Science of the Total Environment
Sustainability
The Cryosphere
The Innovation Geoscience
Water
Water Research
Water Resources Research

Journal Editorial Service

Sustainability
Discover Water
HydroResearch

Other Reviews

National Natural Science Foundation of China Grant Review

Second Order Draft of the IPCC AR6 (group review)

Degree thesis of the Ministry of Education of the People's Republic of China

Organisation of Conferences

Convened the Session BG13 Aquatic Biogeochemical Cycles in Asia Under Climatic and Human Disturbances at the 22nd annual meeting of the Asia Oceania Geosciences Society (AOGS 2025). Singapore, 01/08/2025.

Convened the Session 18.13 Cryosphere Change and Carbon Cycle at the 10th Young Scientist Forum of Earth Science. Hefei, China, 10/05/2025.

Professional Organizations

Members of: American Geophysical Union, Permafrost Young Researchers Network, Chinese Society of Cryospheric Science, Ecological Society of China, Chinese Hydraulic Engineering Society.

Executive Committee of Permafrost Young Researchers Network (PYRN)